ADC 牙科连锁运营管理系统

附图 D-01 / 全系统实体关系图

PK 主键　FK 外键　UQ 独立唯一　? 可空　CF 组合引用（保留列序）

实线为表间引用，箭头指向被引用表；不表示业务流程或多重性。公共及次要引用见蓝色字段目标，可点击定位。

116 TABLES / 768 COLUMNS

285 SINGLE FKs / 21 COMPOSITE FKs

执业关系与专业资质

practice_assignment
跨店执业关系
PK practice_id
FK employee_id -> employee
FK clinic_id -> clinic
valid_from
valid_to ?
practice_arrangement
affected_arrangements ?
FK updated_by -> employee
updated_at
row_version

practice_revision
执业关系原版
PK practice_revision_id
FK practice_id -> practice_assignment
version_no
FK employee_id -> employee
FK clinic_id -> clinic
valid_from
valid_to ?
practice_arrangement
affected_arrangements ?
FK updated_by -> employee
updated_at
FK operation_id -> business_operation

qualification
证书稳定身份
PK qualification_id
FK employee_id -> employee
row_version

qualification_version
牙医证书版本
PK qualification_version_id
FK qualification_id -> qualification
certificate_name
certificate_number ?
source_validity ?
pending_note ?
review_conclusion ?
version_no
FK updated_by -> employee
updated_at

qualification_file
证书附件
PK FK qualification_version_id -> qualification_version
PK FK file_id -> file_asset

员工与技能原版

employee
员工
PK employee_id
name
contact_info ?
personal_profile ?
employment_arrangement ?
FK professional_editor_id ? -> employee
professional_edited_at ?
row_version

employee_revision
员工原版
PK employee_revision_id
FK employee_id -> employee
version_no
name
contact_info ?
personal_profile ?
employment_arrangement ?
FK professional_editor_id ? -> employee
professional_edited_at ?
FK operation_id -> business_operation

employee_skill
员工专业技能
PK FK employee_id -> employee
PK skill_name

employee_revision_skill
员工原版技能
PK FK employee_revision_id -> employee_revision
PK skill_name

诊所与启停原版

clinic
诊所
PK clinic_id
name
address ?
contact_info ?
FK manager_id ? -> employee
area_value ?
area_unit ?
FK created_by -> employee
created_at
FK updated_by -> employee
updated_at
row_version

clinic_revision
诊所原版
PK clinic_revision_id
FK clinic_id -> clinic
version_no
name
address ?
contact_info ?
FK manager_id ? -> employee
area_value ?
area_unit ?
FK created_by -> employee
created_at
FK updated_by -> employee
updated_at
FK operation_id -> business_operation

clinic_availability_plan
诊所业务启停安排
PK availability_id
FK clinic_id -> clinic
effective_from
effective_to ?
business_enabled
reason ?

项目与价格适用范围

treatment_item
治疗项目
PK item_id
name
charge_unit
activation_arrangement
row_version

item_price_version
项目价格版本
PK price_version_id
FK item_id -> treatment_item
amount
effective_from
effective_to ?
version_no

item_price_clinic
价格适用诊所
PK FK price_version_id -> item_price_version
PK FK clinic_id -> clinic

账号、角色与当前授权

staff_account
员工账号
PK account_id
FK UQ employee_id -> employee
UQ login_name
credential_verifier
account_scope ?
activation_arrangement
opened_at
disabled_at ?
FK maintained_by -> employee
maintained_at
row_version

security_role
访问角色
PK role_id
name
responsibility ?
row_version

role_version
角色功能原版
PK role_version_id
FK role_id -> security_role
version_no
name
responsibility ?
FK configured_by -> employee
configured_at

role_function
角色功能
PK FK role_version_id -> role_version
PK function_code

access_grant
账号授权版本
PK grant_id
FK account_id -> staff_account
version_no
effective_from
effective_to ?
object_scope ?
FK configured_by -> employee
configured_at

grant_role
授权采用角色
PK FK grant_id -> access_grant
PK FK role_version_id -> role_version

grant_clinic
授权允许诊所
PK FK grant_id -> access_grant
PK FK clinic_id -> clinic

grant_field
授权字段及动作
PK grant_field_id
FK grant_id -> access_grant
resource_code
field_code
action_code

业务操作与成功命令

business_operation
业务操作历史
PK operation_id
FK actor_id -> employee
FK analysis_id ? -> analysis_run
occurred_at
action_code
subject_kind
subject_key
before_content ?
after_content ?
reason ?
access_scope ?

operation_clinic
操作实际诊所范围
PK FK operation_id -> business_operation
PK FK clinic_id -> clinic

command_dedup
成功命令去重
PK request_key
FK actor_id -> employee
command_code
payload_fingerprint
FK UQ operation_id -> business_operation
result_kind
result_key

经营分析原查询

analysis_run
经营分析查询原版
PK analysis_id
FK employee_id -> employee
analysis_type
period_from
period_to
business_timezone
data_as_of
statistical_rule
completeness_note
queried_at
source_version_manifest
result_snapshot

analysis_scope
分析原诊所范围
PK FK analysis_id -> analysis_run
PK FK clinic_id -> clinic

analysis_staff_scope
分析人员及维度
PK analysis_staff_scope_id
FK analysis_id -> analysis_run
FK employee_id ? -> employee
dimension_definition

供应商与附件

supplier
供应商
PK supplier_id
name
contact_info ?
cooperation_info ?
activation_arrangement
pending_note ?
FK updated_by -> employee
updated_at
row_version

supplier_file
供应商资质合同
PK supplier_file_id
FK supplier_id -> supplier
FK file_id -> file_asset
file_purpose

医生排班与原版

doctor_schedule
医生排班
PK schedule_id
FK doctor_id -> employee
FK clinic_id -> clinic
starts_at
ends_at
can_receive
change_reason ?
row_version

schedule_revision
排班原版
PK schedule_revision_id
FK schedule_id -> doctor_schedule
version_no
FK doctor_id -> employee
FK clinic_id -> clinic
starts_at
ends_at
can_receive
change_reason ?
FK operation_id -> business_operation

预约与真实联系

appointment
就诊预约
PK appointment_id
FK patient_id -> patient
FK clinic_id -> clinic
FK doctor_id -> employee
scheduled_at
cancelled_at ?
confirmation_note ?
row_version

appointment_revision
预约原版
PK appointment_revision_id
FK appointment_id -> appointment
version_no
FK patient_id -> patient
FK clinic_id -> clinic
FK doctor_id -> employee
scheduled_at
cancelled_at ?
confirmation_note ?
FK operation_id -> business_operation

appointment_contact
预约联系历史
PK contact_id
FK appointment_id -> appointment
contacted_at
FK employee_id -> employee
result_note
next_action ?
reached_patient

患者档案与原版集合

patient
患者
PK patient_id
name
birth_date ?
phone ?
contact_address ?
billing_address ?
email ?
source_note ?
introducer_note ?
allergy_knowledge ?
allergy_note ?
contact_preference ?
row_version
FK created_clinic_id ? -> clinic

patient_revision
患者原版
PK patient_revision_id
FK patient_id -> patient
version_no
name
birth_date ?
phone ?
contact_address ?
billing_address ?
email ?
source_note ?
introducer_note ?
allergy_knowledge ?
allergy_note ?
contact_preference ?
FK operation_id -> business_operation
FK created_clinic_id ? -> clinic

patient_tag
患者标签
PK FK patient_id -> patient
PK tag_text

patient_revision_tag
患者原版标签
PK FK patient_revision_id -> patient_revision
PK tag_text

patient_channel
患者沟通渠道
PK channel_id
FK patient_id -> patient
channel_name
association_note ?

patient_revision_channel
患者原版渠道
PK FK patient_revision_id -> patient_revision
PK FK channel_id -> patient_channel
channel_name
association_note ?

patient_allergy
患者过敏明细
PK FK patient_id -> patient
PK item_key
allergen
description ?

patient_revision_allergy
患者原版过敏明细
PK FK patient_revision_id -> patient_revision
PK item_key
allergen
description ?

实际就诊与护士集合

visit
实际就诊
PK visit_id
FK patient_id -> patient
FK clinic_id -> clinic
FK UQ appointment_id ? -> appointment
FK registrar_id -> employee
FK doctor_id -> employee
arrived_at
started_at ?
ended_at ?
left_at ?
row_version
FK appointment_revision_id ? -> appointment_revision
CF01 (appointment_revision_id, appointment_id)
-> appointment_revision (appointment_revision_id, appointment_id)

visit_revision
就诊原版
PK visit_revision_id
FK visit_id -> visit
version_no
FK patient_id -> patient
FK clinic_id -> clinic
FK appointment_id ? -> appointment
FK registrar_id -> employee
FK doctor_id -> employee
arrived_at
started_at ?
ended_at ?
left_at ?
FK operation_id -> business_operation
FK appointment_revision_id ? -> appointment_revision
CF02 (appointment_revision_id, appointment_id)
-> appointment_revision (appointment_revision_id, appointment_id)

visit_nurse
就诊配合护士
PK FK visit_id -> visit
PK FK nurse_id -> employee

visit_revision_nurse
就诊原版护士集合
PK FK visit_revision_id -> visit_revision
PK FK nurse_id -> employee

实际治疗与牙位

treatment
实际治疗
PK treatment_id
FK visit_id -> visit
FK item_id -> treatment_item
FK confirming_doctor_id -> employee
performed_at
row_version

treatment_revision
实际治疗原版
PK treatment_revision_id
FK treatment_id -> treatment
version_no
FK visit_id -> visit
FK item_id -> treatment_item
FK confirming_doctor_id -> employee
performed_at
FK operation_id -> business_operation

treatment_tooth
治疗适用牙位
PK FK treatment_id -> treatment
PK tooth_code

treatment_revision_tooth
治疗原版牙位
PK FK treatment_revision_id -> treatment_revision
PK tooth_code

加工要求、交接与临床核对

lab_order
义齿加工订单
PK lab_order_id
FK treatment_id -> treatment
FK request_doctor_id -> employee
lab_party
progress_pending_note ?
FK updated_by -> employee
updated_at
row_version
FK current_received_handover_id ? -> lab_handover
FK current_review_id ? -> lab_review
has_open_usability_issue
CF104-01 (current_received_handover_id, lab_order_id)
-> lab_handover (handover_id, lab_order_id)
CF104-02 (current_review_id, current_received_handover_id)
-> lab_review (review_id, received_handover_id)

lab_requirement_version
加工要求原版
PK requirement_id
FK lab_order_id -> lab_order
version_no
requirement_text
expected_return_at ?
FK doctor_id -> employee
recorded_at

lab_handover
加工实际交接
PK handover_id
FK lab_order_id -> lab_order
FK requirement_id -> lab_requirement_version
FK employee_id -> employee
direction
occurred_at
handover_content
matches_requirement ?
CF14 (requirement_id, lab_order_id)
-> lab_requirement_version (requirement_id, lab_order_id)

lab_review
加工临床核对
PK review_id
FK requirement_id -> lab_requirement_version
FK received_handover_id -> lab_handover
FK doctor_id -> employee
conclusion
reviewed_at
is_usable
CF15 (received_handover_id, requirement_id)
-> lab_handover (handover_id, requirement_id)

病历与书写医生

clinical_record
病历
PK record_id
FK UQ visit_id -> visit
clinical_at
body
pending_note ?
row_version

clinical_record_revision
病历原版
PK record_revision_id
FK record_id -> clinical_record
version_no
FK visit_id -> visit
clinical_at
body
pending_note ?
FK operation_id -> business_operation

clinical_record_writer
病历书写医生
PK FK record_id -> clinical_record
PK FK doctor_id -> employee

record_revision_writer
病历原版书写医生
PK FK record_revision_id -> clinical_record_revision
PK FK doctor_id -> employee

影像原版与受控文件

clinical_image
诊疗影像
PK image_id
FK visit_id -> visit
FK file_id -> file_asset
FK archived_by -> employee
image_type
formed_at
description ?
row_version

clinical_image_revision
诊疗影像原版
PK image_revision_id
FK image_id -> clinical_image
version_no
FK visit_id -> visit
FK file_id -> file_asset
FK archived_by -> employee
image_type
formed_at
description ?
FK operation_id -> business_operation

file_asset
受控文件元数据
PK file_id
storage_key
media_type
byte_size
content_digest
availability_note ?
FK created_by -> employee
created_at

随访联系与待处理问题

followup_task
治疗随访安排
PK followup_id
FK treatment_id -> treatment
FK plan_version_id ? -> followup_plan_version
FK node_id ? -> followup_plan_node
doctor_instruction ?
FK instruction_doctor_id ? -> employee
planned_at
planned_content
FK responsible_id -> employee
next_action ?
FK updated_by -> employee
updated_at
row_version
CF16 (node_id, plan_version_id)
-> followup_plan_node (node_id, plan_version_id)

followup_contact
随访真实联系
PK followup_contact_id
FK followup_id -> followup_task
FK employee_id -> employee
contacted_at
channel
result_feedback
next_action ?
reached_patient

followup_issue
随访待处理问题
PK followup_issue_id
FK followup_id -> followup_task
problem
FK doctor_id ? -> employee
progress_conclusion ?
resolved_at ?

长期未诊邀约与后续联系

recall_attempt
长期未诊邀约首次尝试
PK recall_id
FK patient_id -> patient
screening_condition
screened_at
last_actual_visit_at ?
FK first_employee_id -> employee
first_contacted_at
first_channel
first_result
patient_intention ?
next_action ?
FK updated_by -> employee
updated_at
row_version
first_reached_patient

recall_clinic
邀约原筛选诊所
PK FK recall_id -> recall_attempt
PK FK clinic_id -> clinic

recall_contact
邀约后续联系
PK recall_contact_id
FK recall_id -> recall_attempt
FK employee_id -> employee
contacted_at
channel
result_intention
next_action ?
reached_patient
stated_intention ?

方案原版与联系节点

followup_plan
随访方案稳定身份
PK plan_id
enabled
row_version

followup_plan_version
随访方案版本
PK plan_version_id
FK plan_id -> followup_plan
FK item_id -> treatment_item
version_no
FK configured_by -> employee
configured_at
effective_from
effective_to ?

followup_plan_clinic
方案适用诊所
PK FK plan_version_id -> followup_plan_version
PK FK clinic_id -> clinic

followup_plan_node
方案联系节点
PK node_id
FK plan_version_id -> followup_plan_version
time_arrangement
contact_content
clinical_purpose

治疗账单、明细与原版

treatment_bill
治疗账单
PK bill_id
FK UQ visit_id -> visit
FK creator_id -> employee
opened_at
row_version

bill_revision
治疗账单原版
PK bill_revision_id
FK bill_id -> treatment_bill
version_no
FK visit_id -> visit
FK creator_id -> employee
opened_at
FK operation_id -> business_operation
FK visit_revision_id -> visit_revision
CF19 (visit_revision_id, visit_id)
-> visit_revision (visit_revision_id, visit_id)

bill_staff
账单相关医护
PK FK bill_id -> treatment_bill
PK FK employee_id -> employee

bill_revision_staff
账单原版相关医护
PK FK bill_revision_id -> bill_revision
PK FK employee_id -> employee

treatment_bill_line
治疗账单项
PK bill_line_id
FK bill_id -> treatment_bill
FK UQ treatment_id -> treatment
FK price_version_id -> item_price_version
quantity
adopted_unit_price

bill_revision_line
账单原版明细
PK revision_line_id
FK bill_revision_id -> bill_revision
FK bill_line_id -> treatment_bill_line
FK treatment_id -> treatment
FK price_version_id -> item_price_version
quantity
adopted_unit_price
FK treatment_revision_id -> treatment_revision
item_name
charge_unit
CF03 (treatment_revision_id, treatment_id)
-> treatment_revision (treatment_revision_id, treatment_id)

真实收款及采用原单版

receipt
统一收款
PK receipt_id
FK UQ bill_id ? -> treatment_bill
FK UQ sale_id ? -> product_sale
FK collector_id -> employee
amount
payment_method
received_at
row_version

receipt_revision
原实收版本
PK receipt_revision_id
FK receipt_id -> receipt
version_no
FK bill_id ? -> treatment_bill
FK sale_id ? -> product_sale
FK collector_id -> employee
amount
payment_method
received_at
FK operation_id -> business_operation
FK bill_revision_id ? -> bill_revision
FK sale_revision_id ? -> sale_revision
CF17 (bill_revision_id, bill_id)
-> bill_revision (bill_revision_id, bill_id)
CF18 (sale_revision_id, sale_id)
-> sale_revision (sale_revision_id, sale_id)

商品销售、交付与原版

product_sale
商品销售
PK sale_id
FK patient_id -> patient
FK clinic_id -> clinic
FK creator_id -> employee
opened_at
row_version

sale_revision
商品销售原版
PK sale_revision_id
FK sale_id -> product_sale
version_no
FK patient_id -> patient
FK clinic_id -> clinic
FK creator_id -> employee
opened_at
FK operation_id -> business_operation

product_sale_line
商品销售项
PK sale_line_id
FK sale_id -> product_sale
FK material_id -> material
quantity
adopted_unit_price

sale_revision_line
商品销售原版明细
PK sale_revision_line_id
FK sale_revision_id -> sale_revision
FK sale_line_id -> product_sale_line
FK material_id -> material
classification
unit
quantity
adopted_unit_price

product_delivery_line
商品交付分配项
PK delivery_line_id
FK sale_line_id -> product_sale_line
FK stock_id -> stock_balance
delivered_quantity
delivered_at

物资目录与库存

material
物资目录
PK material_id
name
classification
unit
specification ?
activation_arrangement
FK updated_by -> employee
updated_at
row_version

stock_balance
诊所物资库存
PK stock_id
FK clinic_id -> clinic
FK material_id -> material
batch_reference ?
expires_on ?
unit
quantity
row_version

入库、领用与原版

stock_receipt
实际入库
PK stock_receipt_id
FK stock_id -> stock_balance
FK supplier_id -> supplier
FK registrar_id -> employee
quantity
unit
received_at
unit_price ?
batch_expiry_note ?
pending_note ?
row_version

material_issue
材料领用出库
PK issue_id
FK stock_id -> stock_balance
FK recipient_id -> employee
FK registrar_id -> employee
quantity
unit
issued_at
cost_unit_price ?
pending_note ?
row_version

issue_revision
原领用版本
PK issue_revision_id
FK issue_id -> material_issue
version_no
FK stock_id -> stock_balance
FK recipient_id -> employee
FK registrar_id -> employee
quantity
unit
issued_at
cost_unit_price ?
pending_note ?
FK operation_id -> business_operation

季度范围与待处理问题

quarter_run
季度核算上下文
PK quarter_id
quarter_label
business_timezone
quarter_end_at
first_full_confirmed_at ?
row_version

quarter_problem
季度范围内问题
PK FK quarter_id -> quarter_run
PK FK issue_id -> data_issue

quarter_scope
季度原纳入诊所
PK FK quarter_id -> quarter_run
PK FK clinic_id -> clinic

quarter_doctor_scope
季度原牙医诊所范围
PK FK quarter_id -> quarter_run
PK FK clinic_id -> clinic
PK FK doctor_id -> employee
CF07 (quarter_id, clinic_id)
-> quarter_scope (quarter_id, clinic_id)

诊所提交与采用来源

quarter_submission
诊所季度资料版本
PK submission_id
FK quarter_id -> quarter_run
FK clinic_id -> clinic
version_no
FK submitted_by -> employee
submitted_at
unresolved_note ?
CF08 (quarter_id, clinic_id)
-> quarter_scope (quarter_id, clinic_id)

submission_receipt
提交原实收依据
PK FK submission_id -> quarter_submission
PK FK receipt_revision_id -> receipt_revision
FK receipt_id -> receipt
CF04 (receipt_revision_id, receipt_id)
-> receipt_revision (receipt_revision_id, receipt_id)

submission_issue
提交原材料依据
PK FK submission_id -> quarter_submission
PK FK issue_revision_id -> issue_revision
FK issue_id -> material_issue
CF05 (issue_revision_id, issue_id)
-> issue_revision (issue_revision_id, issue_id)

submission_problem
提交保留问题
PK FK submission_id -> quarter_submission
PK FK issue_id -> data_issue
adopted_conclusion

submission_expense
提交原费用依据
PK FK submission_id -> quarter_submission
PK FK expense_revision_id -> expense_revision
FK expense_id -> operating_expense
CF06 (expense_revision_id, expense_id)
-> expense_revision (expense_revision_id, expense_id)

费用原版与资料问题

operating_expense
运营费用当前资料
PK expense_id
FK clinic_id -> clinic
FK creator_id -> employee
category
amount
occurred_on
pending_note ?
row_version
voucher_reference ?
voucher_note ?

expense_revision
原费用版本
PK expense_revision_id
FK expense_id -> operating_expense
version_no
FK clinic_id -> clinic
FK creator_id -> employee
category
amount
occurred_on
pending_note ?
FK operation_id -> business_operation
voucher_reference ?
voucher_note ?

data_issue
资料检查问题
PK issue_id
FK clinic_id -> clinic
period_note ?
source_locator
category
severity
responsible_role
handling_note ?
notification_note ?
recheck_conclusion ?
reasonable_blank ?
row_version

issue_action
资料问题处理历史
PK issue_action_id
FK issue_id -> data_issue
FK employee_id -> employee
occurred_at
action_conclusion

盘点及真实库存变动

stocktake
盘点
PK stocktake_id
FK clinic_id -> clinic
FK initiator_id -> employee
scope_note
baseline_at
row_version
processed_at ?

stocktake_line
盘点项
PK stocktake_line_id
FK stocktake_id -> stocktake
FK stock_id -> stock_balance
unit
baseline_quantity
counted_at ?
counted_quantity ?
period_net_movement ?
reason_pending_note ?
actual_adjustment ?
baseline_stock_version

stock_movement
库存变动来源
PK movement_id
FK stock_id -> stock_balance
occurred_at
quantity_delta
FK receipt_id ? -> stock_receipt
FK issue_revision_id ? -> issue_revision
FK delivery_line_id ? -> product_delivery_line
FK stocktake_line_id ? -> stocktake_line
FK operation_id -> business_operation
stock_version_before
stock_version_after

整批结果、分项与核算口径

quarter_result_version
季度整批结果版本
PK result_id
FK quarter_id -> quarter_run
FK policy_version_id -> calculation_policy_version
version_no
FK confirmed_by ? -> employee
confirmed_at ?

result_review
整批复核结论历史
PK review_id
FK result_id -> quarter_result_version
FK employee_id -> employee
conclusion
reviewed_at

result_income_source
牙医实收来源核对
PK income_source_id
FK result_line_id -> quarter_result_line
FK receipt_revision_id -> receipt_revision
adopted_amount
basis

quarter_result_line
牙医诊所结果分项
PK result_line_id
FK result_id -> quarter_result_version
FK doctor_id -> employee
FK clinic_id -> clinic
received_income
material_cost
FK quarter_id -> quarter_run
CF11 (result_id, quarter_id)
-> quarter_result_version (result_id, quarter_id)
CF12 (quarter_id, clinic_id, doctor_id)
-> quarter_doctor_scope (quarter_id, clinic_id, doctor_id)
CF13 (result_id, clinic_id)
-> result_submission (result_id, clinic_id)

result_submission
整批采用提交版
PK FK result_id -> quarter_result_version
PK FK clinic_id -> clinic
FK submission_id -> quarter_submission
FK quarter_id -> quarter_run
CF09 (result_id, quarter_id)
-> quarter_result_version (result_id, quarter_id)
CF10 (submission_id, quarter_id, clinic_id)
-> quarter_submission (submission_id, quarter_id, clinic_id)

result_material_source
材料成本来源核对
PK material_source_id
FK result_line_id -> quarter_result_line
FK issue_revision_id -> issue_revision
adopted_cost
basis

result_unassigned_sale
未归属商品实收来源
PK unassigned_id
FK result_id -> quarter_result_version
FK clinic_id -> clinic
FK receipt_revision_id -> receipt_revision
unassigned_amount

calculation_policy_version
核算口径原版
PK policy_version_id
version_no
business_timezone
income_rule
cost_rule
expense_rule

expense_allocation_line
费用分配项
PK allocation_id
FK result_line_id -> quarter_result_line
FK expense_revision_id -> expense_revision
adopted_amount
basis

111项引用绘实线；其余174项保留字段目标名。精确源列、目标键及类型约束见同源字典。

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浅色区域仅为关系阅读分组，不表示分库或微服务边界。